

Cloud Computing Practical Examination

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Total Marks: 40

Duration: 60 Minutes Instructions

- Perform all tasks on your own cloud virtual machine.
 - You may use AWS EC2, GCP Compute Engine VM, or Azure Virtual Machine.
 - Attach screenshots as evidence of completion.
 - Create a single PDF containing all screenshots and answers.
 - Mention question numbers clearly.
 - Ensure all screenshots clearly show successful execution of commands.
 - Look at the end of the pdf for submission instructions
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Question 1: Docker Installation & Nginx Deployment

Task

1. Install Docker on your cloud virtual machine.
2. Verify Docker installation.
3. Download the Nginx image.
4. Run the Nginx container.
5. Access the Nginx welcome page from your browser.

Required Screenshots

- Docker version output • Docker images output
- Docker ps output showing Nginx running
- Browser showing "Welcome to nginx!"

Hint

Relevant Docker topics:

- Images
- Containers
- Port Mapping
- Container Lifecycle

```
Setting up dnsmasq-base (2.92-1ubuntu0.3) ...
Setting up runc (1.4.0-0ubuntu1) ...
Setting up dns-root-data (2025080400build1) ...
Setting up bridge-utils (1.7.1-4ubuntu3) ...
Setting up containerd (2.2.2-0ubuntu1) ...
Setting up ubuntu-fan (0.12.10) ...
Created symlink '/etc/systemd/system/multi-user.target.wants/ubunfa-can.service' -> '/usr/lib/systemd/system/ubunfa-can.service'.
Setting up docker.io (20.1.3-0ubuntu4.1) ...
Processing triggers for dbus (1.16.2-1ubuntu4) ...
Processing triggers for man-db (2.13.1-1build1) ...
Scanning processes...
Scanning linux images...

running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
ubuntu@ip-172-31-24-100:~$ sudo systemctl start docker
sudo systemctl enable docker
Synchronizing state of docker.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable docker
ubuntu@ip-172-31-24-100:~$ docker --version
Docker version 20.1.1, build 20.1.3-0ubuntu4.1
ubuntu@ip-172-31-24-100:~$
```

Question 2: Docker Volume Practical

Task

1. Create a Docker volume named: studentdata
2. Run an Ubuntu container using the volume.
3. Create a file: /data/info.txt
4. Write your name and roll number inside the file.
5. Display the contents of the file.

Required Screenshots

- Volume creation
- Running container
- File creation
- File contents displayed using cat

- ☐ Docker Volume Creation
 - ☐ File Creation and Verification
 - ☐ Custom Docker Image Build
 - ☐ Custom Docker Container Execution
 - ☐ Bonus Questions (Optional)
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Submission Instructions

After completing the practical examination:

Step 1: Create a PDF

Combine all screenshots and answers into a single PDF document.

Naming Convention

studentname_id_branch.pdf Example:

ImranSayyed_12345_CSE.pdf

Step 2: Repository Information

Repository: <https://github.com/sayyedimran2599/azure-services>

Submission Branch:

practical_exam

Step 3: Switch to Submission Branch

If the branch is not already available locally, fetch the latest branch information from GitHub.

Switch to the branch:

practical_exam

Step 4: Copy Your PDF

Place the completed PDF inside the repository folder.

Verify that the PDF is visible inside the repository.

Step 5: Commit Message Format

Use the following format:

Name_ID_Q1Q2Q3

Examples:

ImranSayyed_12345_Q1Q2Q3

AishwaryaYadav_67890_Q1Q2

RahulPatil_54321_Q1

Where:

Q1Q2Q3 = All questions attempted

Q1Q2 = Questions 1 and 2 attempted

Q1 = Only Question 1 attempted

Step 6: Push Your Submission

Push your changes to the:

“practical_exam” branch.

Step 7: Verify Submission

Verify that:

- Your PDF is present in the repository.
- Your commit message follows the required format.
- The file is visible in the practical_exam branch.

Your submission is considered complete only after the PDF is successfully pushed to GitHub.

